

ARUSH JASUJA

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Authorized to work in the US (F-1 STEM OPT).

SUMMARY

Software engineer with 3 years building and debugging ML infrastructure in production: model deployment, model evaluation, optimization, and data processing for GPU-accelerated deep learning systems serving paying customers. MS Computer Science, NYU. 5 peer-reviewed ML publications.

EDUCATION

New York University | MS, Computer Science | GPA 3.83/4.0 | Sep 2024 - May 2026

Coursework: Graphics Processing Units (Prof. Mohamed Zahran), Deep Learning (Prof. Yann LeCun), Algorithms, Operating Systems

Guru Gobind Singh Indraprastha University | B.Tech, Computer Science | CGPA 8.0/10 | Sep 2016 - Sep 2020

EXPERIENCE

Lead Software Engineer | **Ripik.AI** | **New Delhi, India** | **Aug 2022 - Jul 2024**

- Cut a customer-facing ML planning pipeline from 14 hours to 2 hours for 8 manufacturing facilities. Profiled CNN/ViT inference with Nsight Compute, fixed uncoalesced memory access (AoS to SoA layout), and moved inference to mixed precision (FP16/BF16); GPU throughput improved roughly 40% on the same hardware, deferring a \$15M+ hardware purchase.
- Built multi-node distributed training infrastructure in PyTorch: started on DDP with NCCL, then migrated to FSDP (ZeRO-3) with forward prefetch after tracing AllReduce to 58% of step time. Per-GPU model state dropped 96% (36.8 GB to 1.53 GB across 24 cloud GPUs) and step time fell 27.7%.
- Owned ML infrastructure end to end: model deployment through CI/CD with an 8-minute rollback path, per-class model evaluation gates, and data processing pipelines for sensor and image data. A serving-layer interface I designed cut new-model integration from 3 days to under 4 hours.
- Debugged a silent INT8 quantization regression: rare defect classes lost 12% accuracy, invisible at 0.24% overall. Rolled back in 8 minutes via CI/CD, root-caused calibration range clipping, shipped stratified calibration and per-class accuracy gates.
- Mentored junior engineers on profiling methodology; one engineer's first solo optimization raised a kernel's memory throughput 3.2x against a CI-tracked baseline.

Software Engineer, AI/ML | **Sequoia Insilico** | **New Delhi, India** | **May 2021 - Aug 2022**

- Reduced CNN and LSTM inference latency 45% across 1M+ patient records using PyTorch profiling and INT8 post-training quantization, holding COVID-19 detection accuracy at the 90% clinical floor. Oversampled the rare positive class during calibration so per-class accuracy couldn't drop undetected.

PUBLICATIONS

5 peer-reviewed ML publications across deep learning, computer vision, and medical-imaging ML ([Google Scholar](#)). Two in Alzheimer's & Dementia 2025 (Impact Factor 11.1):

- "[Connecting the Dots with Deep Learning: A Graph-Based Approach of Alzheimer's Conversion Prediction.](#)" *Alzheimer's & Dementia*, 2025.
- "[Layer-wise Adaptive Sine Activation Based Recurrent Network for MCI Conversion.](#)" *Alzheimer's & Dementia*, 2025.

PROJECTS

[Distributed ML Model Serving Platform](#): production Model-as-a-Service on GCP: serves a DistilBERT model on GKE Autopilot, auto-scaling 2-20 replicas at p95 < 150ms and 200+ requests/second, with CI/CD (GitHub Actions), model versioning, Terraform IaC, and Prometheus/Grafana monitoring.

[GPU-Optimized N-Body Simulation](#) (NYU, Prof. Zahran's GPU course), Dec 2025: 6,095x speedup for a 100K-particle simulation on an RTX 4070. Ran roofline analysis before writing any GPU code (1.67 FLOP/byte vs 58 FLOP/byte ridge: memory-bound by 35x); the AoS-to-SoA layout change alone measured 4.26x against a ~3x analytical prediction, with the gap explained by TLB fragmentation. Final kernels hit 60% compute / 52% memory utilization in Nsight Compute. CUDA C++.

SKILLS

Languages: Python, C++ (CUDA C/C++), C, SQL, Bash

ML frameworks: PyTorch, JAX, TensorFlow

ML infrastructure: model deployment, model evaluation, optimization, data processing, debugging, distributed training (DDP, FSDP/ZeRO, NCCL), mixed precision (FP16/BF16), quantization (INT8, FP8), TensorRT, vLLM, SGLang, Triton kernels, CI/CD, Docker, Kubernetes

GPU & Systems: CUDA kernel development, memory-hierarchy optimization, roofline & Speed-of-Light analysis, Nsight Compute/Systems profiling, GPU architecture, large-scale distributed systems, data structures & algorithms, Linux

Data & streaming: recommendation systems, embeddings and retrieval, Apache Kafka, Apache Spark, Hive, Flink, Cassandra, Pinot, streaming and batch data pipelines, large-scale data processing

Leadership: mentored 15+ engineers; speaker at 3 conferences; NYU course section leader (Architecture, Algorithms)